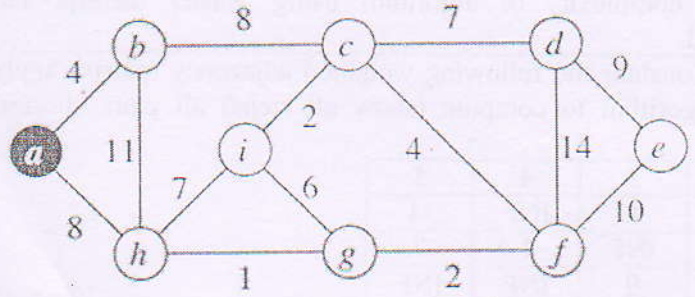
02.05.2025(E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: Computer Engineering.	Class: SY	Semester: IV(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: Computer Engineering.	
Course Code: 116U01C402	Name of the Course: Analysis of Algorithm.	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	List different methods for solving recurrence? Explain any one with example.	5
ii)	Differentiate and discuss with the help of examples? a) Algorithm, b) Procedure and c) Program?	5
iii)	Discuss with the help of example backtracking and, branch and bound problem solving techniques.	5
iv)	Discuss worst case and best case time complexity of insertion sort method with example?	5
v)	Solve following recurrence relation by using Master Method. $T(n) = 2T(n/2) + n$.	5
vi)	Define asymptotic notations. Describe asymptotic notations with the help of graph.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Solve the following recurrence relation $T(n) = T(2n/3) + T(n/3)$ by using recursion tree method.	5
ii)	Solve following recurrence using substitution method. $T(n)=3T(n/3) + n^3$.	5
	OR	
Q2 A	Discuss implementation difference between Kruskal's and Prim's algorithm for Minimum Spanning Tree. Use the Kruskal's and Prim's algorithm and find out the minimum spanning tree. Explain the steps. The source vertex is "a".  	10

Q 2 B	Solve any **One**	10
i)	Solve the 8-puzzle problem for initial and goal state using branch and bound strategy.	10

Initial State			Final State		
1		3	1	2	3
4	2	5	4	5	6
7	8	6	7	8	

ii)	Given two sequences of characters A= "EXPONENTIAL" B="POLYNOMIAL". Obtain longest common subsequence by dynamic approach.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Discuss the process and show the comparisons the KMP string matcher makes for the pattern p="XYXYZXYXY" in the text T="XYXYMXYXZMXYXYZXYXY".	10
ii)	Use dynamic programming algorithm to find an optimal parenthesization of matrix chain multiplication of given sequence of dimension (13, 5, 89, 3, 34).	10
iii)	Consider the following weighted adjacency matrix; find the optimal travelling salesman tour. Assume that starting vertex is 1. Also discuss time complexity of TSP by using dynamic programming approach.	10

	1	2	3	4
1	0	12	17	22
2	5	0	9	10
3	6	13	0	12
4	7	8	9	0

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain sum of subset problem. Apply backtracking to solve the following instance of the subset of sum problem S= {2, 3, 4, 6}, m=11 with the help of state space tree.	10
ii)	Sort the following list of elements in ascending order using Merge Sort 37, 17,22,20,27,13,55,32. Also compute time complexity of algorithm using Master method and Recursion tree method.	10
iii)	For a given graph (Consider the following weighted adjacency matrix) apply Floyd Warshall's Algorithm to compute (show all steps) all pairs shortest distance.	10

	1	2	3	4	5
1	0	3	8	INF	-4
2	INF	0	INF	1	7
3	INF	4	0	INF	INF
4	2	INF	-5	0	-2
5	8	5	1	6	0

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	Discuss difference between NP Complete and NP Hard Problems.	5
ii)	What is NP Completeness and Reducibility?	5
iii)	Discuss difference between class P and class NP	5
iv)	Explain the concept of naive string matching algorithm with the help of suitable example?	5
v)	Compare Bellman ford and Dijkstra's Algorithm with example , also discuss advantages and limitations of both.	5
vi)	Compare divide and conquer approach, and dynamic programming problem solving approach.	5